

TVAC Methodology & Calibration v1

Technical note – Version 1.1 (6 December 2025)

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1. Purpose and scope

The Tokenization Value-Add Calculator (TVAC) is an evaluation engine for real-world asset tokenization cases. It is designed to answer one practical question:

Does this specific tokenization design create net value compared to conventional structures?

This note explains, in non-marketing terms:

- What TVAC looks at
- How the five core factors and the Added Value number are defined
- How verdicts are derived from scores
- How the system is calibrated and kept reasonably stable over time – including checks against both synthetic and historic real-world cases
- What TVAC does *not* do

The note is written for practitioners who want to understand the logic behind the interface: lawyers, product owners, structuring teams, platform operators, and regulators or supervisors.

TVAC is an informational tool. It does **not** replace detailed legal, tax, technical or regulatory advice in any jurisdiction.

TVAC-methodology-v1

2. High-level design

2.1 Conceptual model

TVAC treats each tokenization case as a structured scenario, defined by:

- Asset and legal wrapper
- Jurisdictions (issuer, underlying asset location, and sale markets)

- Investor segments (professional, semi-professional, accredited/qualified, retail)
- Liquidity and venue design
- Custody and settlement model
- Disclosure and regulatory path
- Objectives and constraints (access, cost, risk, compliance, internal policies, tax constraints, etc.)

Each case is evaluated across five core factors:

1. New Opportunities (NO)
2. Cost Savings (CS)
3. Risk Reduction (RR)
4. Tokenization Costs (TC)
5. New Risks (NR)

The three upside factors (NO, CS, RR) represent potential value creation.

The two friction factors (TC, NR) represent extra costs and risks introduced by tokenization.

2.2 Added Value formula

TVAC combines the factor scores into a single Added Value number using:

$$\text{Added Value} = (\text{New Opportunities} + \text{Cost Savings} + \text{Risk Reduction}) - (\text{Tokenization Costs} + \text{New Risks})$$

In shorthand:

- **Plus** = NO + CS + RR
- **Minus** = TC + NR
- **Added Value** = Plus – Minus

Positive values indicate net value creation, negative values indicate net drag, and values around zero indicate borderline cases.

3. Inputs

TVAC uses two main types of input:

1. **Free-text case description**

A short narrative where the user explains:

- what is being tokenized,
- how the structure works,
- who the investors are,
- what the objectives are,
- and any special constraints.

2. **Structured fields**

Key parameters used to anchor the narrative, including:

- Issuance jurisdiction
- Sale jurisdictions
- Investor types
- Instrument type
- Custody model
- Liquidity cadence and venue type
- Settlement model
- Transfer restrictions
- Optional tax notes

The engine does not scrape the web or query hidden external databases at runtime. The assessment is based on the description and structured fields provided by the user.

4. The five factors (0–10 scale)

Each factor is scored from 0 to 10. In broad terms:

- **0–2:** very low impact
- **3–4:** low impact
- **5–6:** moderate / mixed impact
- **7–8:** high impact
- **9–10:** very high impact

For upside factors (NO, CS, RR), higher scores are **better**.

For friction factors (TC, NR), higher scores are **worse**.

Below is an overview of what each factor represents. The internal rubrics that map concrete case features to scores are more granular than this summary, but follow the same logic.

4.1 New Opportunities (NO)

Case-specific upside that is only possible – or clearly strengthened – by tokenization compared to a conventional setup. Examples include:

- Access to new or broader investor groups
- More flexible or realistic secondary liquidity
- Programmability (distributions, conditional cash flows, composability with other infrastructure)
- Better product–market fit that would be difficult with purely off-chain infrastructure

Very low scores indicate that tokenization is essentially cosmetic. High scores indicate that tokenization is a central enabler of the design.

4.2 Cost Savings (CS)

Structural reduction in lifecycle costs vs. a well-run traditional baseline, for example:

- More efficient onboarding and KYC/AML workflows
- Less manual cap-table and record-keeping work
- More automated corporate actions and distributions
- Simpler secondary settlement and fewer intermediaries

Low scores indicate that most traditional cost layers remain in place. Higher scores indicate that multiple lifecycle steps are realistically simplified or automated.

4.3 Risk Reduction (RR)

Improvement in operational, counterparty, governance or transparency risk compared with the baseline, such as:

- Clearer and more auditable records of positions and flows
- Stronger segregation of client assets
- Better delivery-versus-payment mechanics
- More robust governance and oversight structures

Low scores indicate that risks are at best unchanged. Higher scores indicate deliberate use of tokenization to mitigate specific, material risks.

4.4 Tokenization Costs (TC)

Incremental costs introduced by tokenization, beyond what a conventional structure would require, including:

- Additional legal and structuring work
- Platform, custody and venue fees
- Smart-contract development, audits and integrations
- Coordination and change-management overhead

Here, higher scores are worse. Low scores indicate that the cost premium is modest; high scores indicate that tokenization adds substantial cost layers that must be justified by the upside.

4.5 New Risks (NR)

Additional risks created by tokenization vs. the baseline, such as:

- Technology and smart-contract risk

- Cyber and custody risk (private keys, provider breaches, irreversible transfers)
- Regulatory and classification risk under evolving regimes
- Market-structure and liquidity risk on new venues

Again, higher scores are worse. Low scores mean that new risks are tightly managed, well-understood and comparatively small; high scores mean that tokenization adds significant uncertainty or fragility.

5. Scoring and verdicts

5.1 Scoring workflow (high level)

At a high level, the evaluation follows this pattern:

1. Interpret the case based on the free-text description and structured fields.
2. Identify mechanisms relevant to each factor (what drives new opportunities, where costs and risks arise, etc.).
3. Assign factor scores on a 0–10 scale using internal rubrics.
4. Compute Plus, Minus and Added Value using the formula.
5. Derive a verdict band (Viable, Borderline, Non-viable, or Conditional Go) based on the Added Value range and the internal consistency of the design.
6. Generate a structured report (summary, factor explanations, value pathways, risk register, design suggestions and an indicative roadmap).

5.2 Verdict bands (indicative)

While exact thresholds may change slightly between versions, the logic is:

- **Viable / Go**
Added Value clearly positive, no obvious structural blockers, and

a coherent story where upside outweighs added cost and risk.

- **Borderline / Mixed**

Added Value modestly positive or close to zero; several open questions or dependencies; case could become clearly viable or non-viable depending on further design decisions.

- **Non-viable / No-Go**

Added Value clearly negative or presence of obvious structural blockers (e.g. banned activity, impossible regulatory path, incompatible custody/venue/legal combination).

- **Conditional Go**

Conceptually viable and potentially attractive, but dependent on specific conditions being met (e.g. successful resolution of legal, tax or platform dependencies).

TVAC is intentionally somewhat conservative: when a case sits on the boundary, the more cautious interpretation is usually chosen.

5.3 Input completeness and assumptions

TVAC tracks a simple notion of input completeness:

- A numeric score indicating how coherent and complete the input appears.
- A list of missing fields and explicit assumptions the evaluator relied on.

Thin or contradictory inputs should not be treated the same as a carefully specified case. Users are encouraged to review and, where necessary, refine their cases based on the “missing fields” and “assumptions” sections.

A more explicit uncertainty score is planned for future versions.

6. Calibration and reference cases

6.1 Calibration philosophy

TVAC is **not** re-optimised dynamically per user. The scoring framework is:

- Defined as a fixed structure (factors, formula, verdict bands)
- Calibrated against a small set of synthetic reference cases *and* a sample of historic, named tokenization projects
- Checked for consistency rather than “perfect accuracy” on any single case

The objective is a stable, explainable profile that is sceptical of over-optimistic claims and helpful as a structured input to human decision-making – not a back-test of marketing decks for individual transactions.

6.2 Internal synthetic reference cases

Internally, TVAC is first calibrated on a library of synthetic cases that represent typical design patterns, for example:

- Simple, professional-only issues in a single jurisdiction
- More complex cross-border and semi-retail structures
- Infrastructure-style use cases focused on operational efficiency and risk reduction
- Cases where tokenization adds little beyond a cosmetic wrapper

For each reference case, there is an expected *shape* of the profile (relative levels of NO, CS, RR, TC, NR and the likely verdict band). The internal rubrics are adjusted so that TVAC produces profiles consistent with these expectations.

The detailed synthetic reference set and exact parameter values are kept internal, to avoid turning this note into a step-by-step

implementation guide. The goal is transparency about the *approach*, not publication of every threshold.

6.3 Historic real-world cases (sanity checks)

In addition to synthetic cases, TVAC has been sanity-checked against a small set of well-known tokenization projects from the last decade.

These include, for example:

- Early tokenized VC and private-markets funds
- Regulated SME bond and note offerings
- Early US and European equity-style STOs on ATS/MTF venues
- Regulated issues on Asian digital-securities exchanges
- Early European real-estate tokenizations

For each of these historic cases, the public structure (jurisdictions, investor focus, venue type, custody model, etc.) is re-interpreted in neutral language and entered into TVAC. We then compare:

- the qualitative verdict band (Viable / Borderline / Non-viable / Conditional Go), and
- the overall Added Value level and balance between upside and friction factors,

against an internal prior view based on independent research and expert judgement.

In the current calibration:

- Verdicts on the historic cases align with our prior qualitative assessment in practice – TVAC does **not** produce obviously contradictory “go/no-go” signals for these transactions.
- Differences at factor level are within a narrow band and mostly reflect TVAC’s intentionally cautious stance (for example scoring New Risks slightly higher than some original marketing material would suggest).

- The rubrics have **not** been micro-tuned to force perfect numerical matches; where small discrepancies exist, they are accepted as part of model uncertainty.

These checks are meant as **sanity checks**, not as claims about the commercial success or failure of specific historic offerings.

6.4 Regression checks

Whenever the evaluator is materially updated (for example, changes to prompts or underlying model versions), the internal synthetic and historic reference cases are re-run and compared with previous results inside broad tolerance bands.

The focus is on:

- preserving the overall shape of results,
- avoiding unexplained verdict flips for unchanged cases, and
- keeping the conservative stance intact.

7. Limitations and appropriate use

Key limitations:

1. **No live regulatory or case-law database**
TVAC does not automatically track every national implementation, case or guidance document.
2. **No guarantee of compliance**
The tool does not replace formal legal, tax, regulatory or technical advice.
3. **Input dependence**
Poor or incomplete descriptions produce speculative assessments; TVAC highlights missing information but cannot fix it.

4. **Model dependence**

The evaluator uses modern language models; while calibrated, they can still over- or under-estimate aspects of a case.

5. **No pricing or return modelling**

TVAC does not forecast returns, default probabilities or price paths; it focuses on structural value-add and feasibility.

Appropriate uses include:

- Early-stage feasibility checks and design discussions
- Internal workshops across business, legal, compliance and technology
- Comparing alternative design options for the same project

It should **not** be used as:

- The sole basis for go/no-go decisions
- A substitute for professional legal, regulatory, tax or investment advice

8. Interpreting a TVAC report

A typical report includes:

- Case summary (verdict, Added Value overview, input completeness)
- Asset-class context (how similar cases tend to behave)
- Factor scores and explanations
- Value pathways and risk register
- Design recommendations and what-if scenarios
- Indicative implementation roadmap

The report is best used as a structured conversation partner and as documentation of how a particular tokenization idea looked at a specific point in time, **not** as a final verdict.

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Disclaimer. *This tool provides informational analysis based on the inputs you provide and publicly available best-practice rubrics. It is not legal, financial, investment or tax advice. Outcomes depend on jurisdiction-specific rules, counterparty requirements, and implementation choices which must be verified with qualified professionals.*

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